

Mistral Releases Leanstral: An Open-Source Foundation for Trustworthy Vibe-Coding

In a major step towards verifiable AI coding, Mistral AI has announced the release of **Leanstral**, the first open-source code agent designed specifically for the Lean 4 proof assistant. This new model aims to address the scaling bottleneck of human review in high-stakes domains by formally proving its implementations against strict specifications.

Verifiable Code Generation

While AI agents have become highly capable at generating code, verifying that code for mission-critical applications still requires extensive human expertise. Leanstral tackles this by integrating with Lean 4, a proof assistant capable of expressing complex mathematical objects and software specifications. This allows the agent not only to write code but to mathematically prove its correctness.

Efficient and Accessible

With 6B active parameters and a highly sparse architecture, Leanstral is optimized for proof engineering tasks. Despite its smaller size, it demonstrates a significant efficiency advantage over much larger open-source models and offers competitive performance against leading proprietary models at a fraction of the cost.

Mistral has released Leanstral's weights under an Apache 2.0 license. It is also available via a free API endpoint and seamlessly integrated into Mistral Vibe for zero-setup vibe coding and proving.

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